Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Department of AIDS

Computer Network
Sub Code: 303105255

Study Guide

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UNIT 1: DATA COMMUNICATION COMPONENTS

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1. Representation of Data

Data communication refers to the exchange of data between two devices through a transmission medium such as wires or wireless channels. Data can be represented in several forms:

Types of Data Representation

Data Type	Representation Method
Text	ASCII (American Standard Code for Information Interchange), Unicode
Numbers	Binary format (0s and 1s), BCD
Images	Pixels represented in binary
Audio	Analog signals; digitized using PCM
Video	Combination of audio + images in binary; formats like MPEG, MP4

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2. Flow of Data

Data travels between communicating devices in different modes:

Data Flow Modes

Flow Mode	Description	Example
Simplex	One-way communication	Keyboard to CPU
Half-Duplex	Two-way communication, but not simultaneously	Walkie-talkie
Full-Duplex	Two-way simultaneous	Telephone

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3. Computer Networks

A network is a collection of connected computers and devices that share resources and data.

Types of Networks

Network Type	Coverage Area	Example
LAN	Local – office, school	Campus LAN
MAN	Metropolitan city area	Cable TV network
WAN	Country / world level	Internet

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4. Connection Topologies

Topology refers to the physical or logical arrangement of network devices.

Types of Topologies

Topology	Description	Advantages	Disadvantages
Bus	Single central cable shared by nodes	Easy and cheap	Fault affects entire network
Star	All devices connected to central switch	Easy management	Failure of hub = failure of network
Ring	Devices form a circular path	High speed	Failure of single node breaks network
Mesh	Fully connected devices	Reliable	Expensive and complex
Hybrid	Combination of topologies	Flexible	Complex design

5. Protocols and Standards

Protocols are rules governing communication. Standards ensure compatibility among devices.

Category	Examples
Protocol Examples	TCP/IP, FTP, HTTP, SMTP, POP3
Standards Organizations	IEEE, ISO, ITU-T, ANSI
Popular Standard Models	IEEE 802 standards (LAN/WLAN), ISO OSI model

6. OSI Reference Model

The Open System Interconnection model defines 7 layers involved in network communication.

Layer No.	Name	Function
7	**Application**	Services for user applications (HTTP, FTP)
6	**Presentation**	Data translation, encryption, compression
5	**Session**	Establishes, manages, and terminates sessions
4	**Transport**	End-to-end delivery, segmentation (TCP/UDP)
3	**Network**	Routing of packets (IP)
2	**Data Link**	Framing, error detection, MAC addressing
1	**Physical**	Transmission media, signals, data rate

7. Transmission Media

Divided into **guided (wired)** and **unguided (wireless)**.

Guided Media

Type	Description	Example Speed
Twisted Pair	Telephone & LAN cables	10 Mbps – 1 Gbps
Coaxial	Cable TV	Up to 10 Gbps
Fiber Optic	Light pulses	Highest speed, 100 Gbps+

Unguided Media

Type	Example
Radio Waves	FM radio, WiFi
Microwaves	Satellite
Infrared	Remote controls

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8. Local Area Networks (LAN)

Wired LAN

- Uses Ethernet cables (UTP, STP)
- Devices connected via switches/routers
- IEEE 802.3 standard

Wireless LAN (WLAN)

- Uses Wi-Fi technology
- Devices connect via access point
- IEEE 802.11 a/b/g/n/ac/ax standards

Connecting LANs

Device	Function
Hub	Broadcasts data
Switch	Sends data to specific MAC address
Router	Connects multiple networks (LAN-WAN)

Virtual LAN (VLAN)

- Logical segmentation of LAN using switches
- Separates traffic for security & performance
- IEEE 802.1Q standard

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9. Bandwidth Utilization Techniques

Used to increase data-carrying capacity.

Multiplexing

Type	Description	Example
Frequency Division Multiplexing (FDM)	Multiple signals, each with different frequency	Radio channels
Time Division Multiplexing (TDM)	Time is divided into slots	Digital telephony

Type	Description	Example
Wavelength Division Multiplexing (WDM)	Light wavelengths multiplexed in fiber optics	Long-distance fiber networks

10. Spread Spectrum Techniques

Used in wireless communication for security and interference reduction.

Type	Description
FHSS (Frequency Hopping Spread Spectrum)	Rapidly jumps among frequencies
DSSS (Direct Sequence Spread Spectrum)	Spreads signal over wide frequencies

Used in: Bluetooth, Wi-Fi, GPS.

UNIT 2: DATA LINK LAYER & MAC SUBLAYER

1. Error Detection and Error Correction

Fundamentals

- During data transmission over communication channels, noise can introduce **errors** causing **bit alteration** (0→1 or 1→0).
- **Error detection** identifies the presence of errors.
- **Error correction** involves identifying and correcting errors.
- Types of errors:
  - **Single-bit error** – only one bit changes.
  - **Burst error** – two or more consecutive bits change.

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2. Block Coding

- Data is divided into **blocks** of fixed size k bits and encoded to n bits where n > k.
- Extra bits added = **redundant bits (r = n − k)**.
- The receiver checks the received codeword for validity using a decoder.
- A block code is valid when it matches one of the **legal codewords** stored in the decoder.

Example

If k = 4 bits and r = 3 bits → n = 7 bit codeword.

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3. Hamming Distance

- **Hamming distance**: number of different bits between two binary strings.

Example

10101
11100
→ Hamming distance = 2 + 0 + 1 + 1 + 1 = 4

Minimum Hamming Distance (dmin)

- Minimum distance among all codewords.
- To detect d errors → dmin ≥ d + 1
- To correct t errors → dmin ≥ 2t + 1

Hamming Code

- Error-correcting code that corrects **single-bit errors**.
- Redundant bits must satisfy:
- 2^r ≥ m + r + 1

where m = data bits, r = redundant bits

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4. Cyclic Redundancy Check (CRC)

- Most powerful error-detecting technique.
- Treats data as a polynomial.
- Divides data polynomial by **generator polynomial (G(x))** using binary division.
- The remainder is called **CRC bits**.
- Sent frame = Data + CRC

Advantages

- Detects burst errors up to degree(G).
- High reliability, easy hardware implementation.

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5. Flow Control & Error Control Protocols

Flow Control

- Prevents sender from overwhelming receiver.
- Maintains data transfer rate.

Error Control

- Uses **ARQ (Automatic Repeat Request)**.

- Combines **error detection + retransmission**.

Stop-and-Wait ARQ
- Sender transmits one frame and waits for ACK before sending next.
- Simple but inefficient due to idle waiting time.

Frame → | wait for ACK | → Next Frame

Drawbacks
- Low efficiency in long-distance networks.

Go-Back-N ARQ
- Uses a **window size N**.
- Sender sends multiple frames sequentially without waiting for ACK.
- If a frame is lost/errored → retransmit **that frame and all following frames**.

Efficiency

Higher throughput but more retransmissions.

Selective Repeat ARQ
- Only the **erroneous frame** is retransmitted.
- Receiver accepts out-of-order frames and stores them until correct frame arrives.

Benefits
- Much more efficient than Go-Back-N.
- Reduces retransmission overhead.

Sliding Window Protocol
- Used for both flow and error control.
- Sender and receiver maintain windows to track sent/received frames.
- Ensures continuous transmission without waiting.

Piggybacking
- In two-way communication, ACK is attached with outgoing frame rather than sending separately.
- Reduces overhead.

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6. Random Access & Multiple Access Protocols

Used when many stations share a common channel (like LAN).

Pure ALOHA
- Send data whenever ready.
- If collision occurs → retransmit after random time.

Throughput

Maximum = **18% (0.18)** efficiency.

Slotted ALOHA
- Time divided into slots.
- Transmission allowed only at the beginning of slots.

Throughput

Maximum = **37% (0.37)** efficiency.

CSMA (Carrier Sense Multiple Access)
- Sense the channel before sending.
- **Persistent** vs **Non-persistent** vs **p-persistent**.

CSMA/CD (Collision Detection)
- Used in Ethernet (wired).
- Detects collisions and **aborts** transmission immediately (jam signal sent).

Steps

1. Sense channel
2. Transmit
3. Collision detect
4. Abort and backoff

CSMA/CA (Collision Avoidance)

- Used in Wi-Fi (Wireless).
- Collisions cannot be detected physically.
- Uses **RTS/CTS (Request to Send / Clear to Send)** to avoid collisions.

CDMA (Code Division Multiple Access)

- Each station uses a unique **Orthogonal code**.
- All users can send simultaneously at full bandwidth.
- Used in cellular mobile communication.

Summary Table

Protocol/Concept	Key Use	Pros	Cons
Stop-and-Wait	Simple ARQ	Easy	Low efficiency
Go-Back-N	ARQ + window	Good throughput	Retransmits many frames
Selective Repeat	Efficient ARQ	Individual frame resend	Complex receiver
Pure ALOHA	Random access	Simple	Low efficiency
Slotted ALOHA	Random access	Better efficiency	Needs synchronization
CSMA/CD	Wired LAN	Fast collision handling	No use in wireless
CSMA/CA	Wireless	Collision avoidance	Delay due to RTS/CTS

UNIT 3: NETWORK LAYER

1. Switching
Switching is the process of moving data packets from the source to the destination across a network. It determines the path that a packet takes.
Types of Switching

Type	Description	Advantages	Disadvantages
Circuit Switching	A dedicated connection is established between source & destination before data transfer.	Fixed bandwidth, reliable, fixed delay.	Inefficient resource usage if line is idle, setup delay.
Packet Switching	Data is divided into packets; each packet travels independently.	Efficient resource use, high reliability.	Variable delay, out-of-order packets.
Message Switching	Entire message is treated as a single unit; stored completely at intermediate nodes (**Store & Forward**).	No size limit, efficient.	Long delay if big message.

Packet Switching Types
- **Datagram packet switching** – packets take different routes; reordering required.
- **Virtual circuit switching** – path predetermined before communication (VPNs).

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2. Logical Addressing
Logical addressing is the process of assigning IP addresses that uniquely identify devices in an internetwork. Implemented by the **Network Layer (Layer 3)**.
IPv4 Addressing
- **32-bit address**, represented in dotted decimal format (e.g., 192.168.10.5)
- Supports about **4.3 billion addresses**
- Written as: **Network ID + Host ID**
- Example: 192.168.1.10/24

IPv4 Address Classes

Class	Range	Default Mask	Usage
A	0–127	255.0.0.0	Large networks
B	128–191	255.255.0.0	Medium networks
C	192–223	255.255.255.0	Small networks
D	224–239	—	Multicasting
E	240–255	—	Research

IPv6 Addressing
- **128-bit** address, written in hexadecimal separated by colons
- Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- Provides **3.4 × 10^38 addresses**

IPv6 Features
- Auto-configuration
- Simpler header compared to IPv4
- Built-in security (**IPSec**)

- No NAT required
- Multicast supported, broadcast removed

IPv6 Address Types

Type	Purpose
Unicast	One-to-one communication
Anycast	Nearest node communication
Multicast	One-to-many communication

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3. Address Mapping

Mapping logical address (IP) to physical address (MAC) and vice-versa.

Address Resolution Protocol (ARP)

- Used to map **IPv4 address → MAC address**
- Source broadcasts ARP request; target responds with ARP reply.

Reverse ARP (RARP)

- Used to map **MAC → IP**
- Used by diskless workstations to obtain IP address.

BOOTP (Bootstrap Protocol)

- Replaced RARP; assigns IP through a server
- Provides IP, gateway, subnet mask, boot file location

DHCP (Dynamic Host Configuration Protocol)

- Assigns IP addresses dynamically
- Supports **automatic, dynamic & manual allocation**

DHCP Message Exchange

Message	Purpose
DISCOVER	Client finds DHCP server
OFFER	Server offers IP
REQUEST	Client requests that offer
ACK	Server confirms and allocates IP

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4. Delivery

Specifies how packets are delivered end-to-end.

Delivery Types

Type	Description
Direct delivery	Source and destination are in the **same network**
Indirect delivery	Uses **routers** when destination is remote

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5. Forwarding

Forwarding determines which outgoing interface a packet should use. Packets are forwarded based on routing table entries.

Forwarding Techniques

Technique	Description
Next-hop	Table stores next router instead of final destination
Network-specific	One entry per network
Host-specific	One entry per host
Default route	Used when no specific route exists

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6. Unicast Routing Protocols

Routing selects the **best path** for packet forwarding.

Types of Routing

Type	Description
Static Routing	Manually configured, no automatic updates
Dynamic Routing	Routers update information automatically

Routing Algorithms

Category	Examples	Description
Distance Vector	**RIP**	Uses hop count as metric
Link State	**OSPF, IS-IS**	Global knowledge of network
Path Vector	**BGP**	Used between autonomous systems

Popular Unicast Routing Protocols

Protocol	Type	Metric	Notes
RIP (Routing Information Protocol)	Distance Vector	Hop count (max 15 hops)	Simple, slow convergence
OSPF (Open Shortest Path First)	Link State	Cost (bandwidth)	Faster convergence, large networks
EIGRP (Enhanced Interior Gateway Routing Protocol)	Hybrid	Bandwidth, delay	Cisco proprietary
BGP (Border Gateway Protocol)	Path Vector	Policy-based	Internet backbone routing

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UNIT 4: TRANSPORT LAYER

Process-to-Process Communication
Process-to-process communication refers to the exchange of data between running applications (processes) on the same or different networked systems.
Key Concepts

Term	Description
Process	A running application or program.
Ports	Logical communication endpoints used to identify specific processes.
Client–Server Model	Client requests a service; server responds.
Socket	Software interface enabling two processes to communicate via network.
IP Address	Identifies a host in a network.
Port Number	Identifies an application/process on the host.

Socket Communication Steps
1. Server creates a socket and binds it to a port.
2. Server listens for client requests.
3. Client creates a socket and connects to server IP and port.
4. Data is exchanged across the network.
5. Connection is terminated after communication ends.

User Datagram Protocol (UDP)
UDP is a **connectionless**, **unreliable**, lightweight transport layer protocol used for fast data transmission.
Features
- **Connectionless** (no setup/handshake)
- **Unreliable** (no acknowledgment, no retransmission)
- **No flow or congestion control**
- Uses **checksum** for minimal error detection
- Supports one-to-one, one-to-many, many-to-one communication

UDP Header Format (8 Bytes)

Field	Size
Source Port	2 bytes
Destination Port	2 bytes
Length	2 bytes
Checksum	2 bytes

Advantages
- Faster than TCP
- Low latency

Applications
- Video streaming, VoIP

- DNS, DHCP
- Online gaming

Transmission Control Protocol (TCP)
TCP is a **connection-oriented**, **reliable** transport protocol that guarantees ordered and error-free delivery.
Features
- Reliable with acknowledgment and retransmission
- Uses **sequence numbers** and **ack numbers**
- Provides **flow control (Sliding Window)**
- **Congestion control**
- Ensures **in-order delivery**
- **Connection-oriented (3-way handshake)**

TCP 3-Way Handshake

Step	Message	Purpose
1	SYN	Client requests connection
2	SYN+ACK	Server accepts connection
3	ACK	Client confirms connection

TCP Header Fields
Sequence Number, Acknowledgment Number, Flags (SYN, ACK, FIN), Window size, Checksum, Urgent Pointer, Options.
Applications
- Web browsing (HTTP/HTTPS)
- Email (SMTP, IMAP)
- File transfer (FTP)

SCTP (Stream Control Transmission Protocol)
SCTP is a **message-oriented**, **reliable** transport protocol designed for telecommunication signaling.
Features
- Supports **multistreaming** (multiple independent streams)
- **Multihoming** (multiple IP addresses per endpoint)
- Reliable, ordered/unordered delivery
- Congestion control and flow control
- 4-way handshake (INIT, INIT-ACK, COOKIE-ECHO, COOKIE-ACK)

Applications
- Telecom signaling (SS7)
- VoIP signaling (SIP)
- Real-time multimedia

Congestion Control
Congestion occurs when the network is overloaded, causing delay or packet loss. **TCP** uses algorithms to handle congestion.
TCP Congestion Control Mechanisms

Mechanism	Description
Slow Start	Begins with a small congestion window, doubles every RTT
Congestion Avoidance	Linear growth after threshold
Fast Retransmit	Retransmits after receiving 3 duplicate ACKs

Mechanism	Description
Fast Recovery	Avoids going back to slow start after packet loss

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Quality of Service (QoS)
QoS refers to techniques to maintain consistent performance for network services.
QoS Parameters

Parameter	Meaning
Bandwidth	Amount of data transmitted per second
Latency	Delay in packet delivery
Jitter	Variation in packet delay
Packet Loss	Dropped packets during transfer
Reliability	Ability to maintain stable communication

QoS Requirements Example
- VoIP → low delay & jitter
- Video streaming → high bandwidth
- File transfer → reliability

QoS Improvement Techniques

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1. Leaky Bucket Algorithm
Used for **traffic shaping** & **congestion control** by sending packets at a **constant rate**.
Working
- Packets enter a bucket with fixed capacity.
- They leave the bucket at a constant rate.
- If bucket is full → excess packets are discarded.

Characteristics
- Controls output rate
- Prevents burst traffic
- Simple & predictable

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2. Token Bucket Algorithm
Allows burst traffic while maintaining a limit.
Working
- Tokens are generated at a fixed rate and stored in a bucket.
- Each packet requires a token to be transmitted.
- If tokens available → packets can be sent immediately
- Allows bursts if tokens accumulated

Comparison

Leaky Bucket	Token Bucket
Fixed output rate	Allows burst transmission
Discards packets if full	Holds packets until tokens available

Leaky Bucket	Token Bucket
More rigid	More flexible and widely used

Summary

Protocol	Type	Reliability	Speed	Use
UDP	Connectionless	No	Fast	Streaming, gaming
TCP	Connection-oriented	Yes	Slower	Web, email, file transfer
SCTP	Message-oriented	Yes	Medium	Telecom signaling

UNIT 5: APPLICATION LAYER

1. Domain Name Space (DNS)

- **DNS stands for Domain Name System/Service/Server.**
- **It is a distributed hierarchical naming system that translates human-readable domain names into IP addresses.**
- **Example: www.google.com → 142.250.182.14**
- **Without DNS, users would need to remember numeric IP addresses.**

Features

- **Distributed database**
- **Hierarchical tree structure**
- **Uses UDP port 53**
- **Caching for faster response**

Hierarchy of Domain Name Space

1. **Root Domain – represented by a dot .**
2. **Top Level Domain (TLD) – edu, com, org, gov, in, uk, etc.**
3. **Second Level Domain – google.com, yahoo.com, etc.**
4. **Subdomain / Host – mail.google.com**

DNS Servers

- **Root Server**
- **TLD Server**
- **Authoritative Server**
- **Local/Resolver**

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2. Dynamic DNS (DDNS)

- **DDNS automatically updates DNS records when the IP address of a device changes.**
- **Essential when using dynamic IPs provided by ISPs.**
- **Used in CCTV remote access, home servers, remote login.**
- **Tools include No-IP, DynDNS, Cloudflare DDNS.**

Working

1. **Client IP changes**
2. **DDNS software detects change**
3. **Updates DNS record automatically**
4. **Users still access same domain name**

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3. TELNET

- **Terminal Network Protocol used for remote login to another computer.**
- **Allows command execution on a remote machine.**
- **Uses TCP port 23**
- **Transfers data in plaintext → not secure.**

Uses

- **Remote server administration**
- **Testing ports and services**
- **Network troubleshooting**

Limitation

- **No encryption → replaced mostly by SSH (Secure Shell).**

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4. EMAIL (Electronic Mail)

- **A system for exchanging text, files, images, and documents electronically.**

Protocols Used

Protocol	Function	Port
SMTP	Send mail	25 / 587

Protocol	Function	Port
POP3	Receive/download mail	110
IMAP	Access mail on server	143

Email Components

- **Email ID (username@domain)**
- **CC, BCC, Subject, Body**
- **Attachments**

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5. File Transfer Protocol (FTP)

- **Protocol used to transfer files between computers.**
- **Uses TCP ports 20 & 21**
- **Supports anonymous access and username/password-based access.**

Features

- **Upload, download, rename, delete files**
- **Supports large files**
- **Can run in Active or Passive mode**

Secure Alternative

- **SFTP (SSH File Transfer Protocol)**
- **FTPS (FTP + SSL/TLS)**

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6. WWW (World Wide Web)

- **System of interlinked hypertext documents accessible through the internet.**
- **Created by Tim Berners-Lee (1991).**
- **Accessed using a web browser (Chrome, Firefox, Safari).**

Components

- **Web pages (HTML)**
- **Web server**
- **Web browser**
- **URL (Uniform Resource Locator)**

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7. HTTP (HyperText Transfer Protocol)

- **Protocol for transferring web documents across the WWW.**
- **Stateless, text-based protocol**
- **Uses TCP port 80**
- **Works on Client-Server architecture**

Versions

- **HTTP/1.1 – persistent connection**
- **HTTP/2 – faster, multiplexing**
- **HTTPS – secure communication using SSL/TLS on port 443**

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8. SNMP (Simple Network Management Protocol)

- **Used for monitoring and managing network devices.**
- **Devices include routers, switches, printers, servers.**
- **Uses UDP ports 161 & 162**

Components

- **Manager – controller system**
- **Agent – installed on devices**
- **MIB (Management Information Base) – database of network objects**

Functions

- **Collect network performance data**
- **Fault detection**
- **Alerts and notifications**

9. Bluetooth

- **Short-range wireless communication technology.**
- **Uses 2.4 GHz ISM band**
- **Range: 10 m – 100 m depending on class.**

Applications

- **File transfer between devices**
- **Wireless headphones, keyboards, speakers**
- **IoT devices, smartwatches**

Advantages

- **Low power consumption**
- **Cost-effective**
- **Easy setup**

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10. Firewalls

- **A network security system that monitors and controls incoming/outgoing network traffic.**
- **Acts as a barrier between trusted internal network and untrusted external network like the Internet.**

Types

- **Packet Filtering Firewall**
- **Stateful Inspection Firewall**
- **Proxy Firewall**
- **Next-Generation Firewall (NGFW)**

Functions

- **Block unauthorized access**
- **Protect against attacks**
- **Provide logging & monitoring**

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11. Basic Concepts of Cryptography

- **Cryptography is the science of securing information by converting it into unreadable form.**
- **Process: Encryption (plaintext → ciphertext) and Decryption (ciphertext → plaintext).**

Types of Cryptography

Type	Key Used	Example
Symmetric	Same key	AES, DES
Asymmetric	Public & Private key	RSA, ECC
Hashing	No key	MD5, SHA-256

Applications

- **Online banking**
- **Digital signatures & certificates**
- **Secure communication**
- **Password protection**

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